Methods of Solving First-Order Partial Differential Equations

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October 26, 2026

First-Order PDEs: Definitions and Notation

Recall

In general, a first-order PDE can be written as:

$$F(x, y, z, p, q) = 0 \quad \text{— (0)}$$

where $z = z(x, y)$ is the unknown function of two independent variables x and y.

Standard Notations:

- $p = \dfrac{\partial z}{\partial x} = z_x$
- $q = \dfrac{\partial z}{\partial y} = z_y$
- $r = \dfrac{\partial^2 z}{\partial x^2} = z_{xx}$
- $s = \dfrac{\partial^2 z}{\partial x \partial y} = z_{xy}$
- $t = \dfrac{\partial^2 z}{\partial y^2} = z_{yy}$

Formation of PDEs

Partial differential equations can formed either by the elimination of arbitrary constants or by the elimination of arbitrary functions from a relation involving three or more variables.

1. By elimination of Constants

Given a relation:

$$F(x, y, z, a, b) = 0 \quad \text{— (1)}$$

where a and b are arbitrary constants. We have to eliminate a and b while forming the differential equation.

Differentiating $F(x, y, z, a, b) = 0$ partially with respect to x:

$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial z}\frac{\partial z}{\partial x} = 0 \implies \frac{\partial F}{\partial x} + p\frac{\partial F}{\partial z} = 0 \quad \text{— (2)}$$

Differentiating partially with respect to y:

$$\frac{\partial F}{\partial y} + \frac{\partial F}{\partial z}\frac{\partial z}{\partial y} = 0 \implies \frac{\partial F}{\partial y} + q\frac{\partial F}{\partial z} = 0 \quad -- (3)$$

By eliminating the two arbitrary constants "a" and "b" from equations (1), (2), and (3), we obtain a partial differential equation of the first order of the form:

$$\Phi(x, y, z, p, q) = 0 \quad -- (4)$$

Example

Question: Form the PDE by eliminating the arbitrary constants a and b from:

$$z = (x^2 + a)(y^2 + b)$$

Solution: Differentiating with respect to x:

$$\frac{\partial z}{\partial x} = 2x(y^2 + b) \implies p = 2x(y^2 + b) \implies y^2 + b = \frac{p}{2x}$$

Differentiating with respect to y:

$$\frac{\partial z}{\partial y} = 2y(x^2 + a) \implies q = 2y(x^2 + a) \implies x^2 + a = \frac{q}{2y}$$

Substituting these into the original equation $z = (x^2 + a)(y^2 + b)$:

$$z = \left(\frac{q}{2y}\right)\left(\frac{p}{2x}\right) \implies 4xyz = pq$$

2. By Elimination of Arbitrary Functions: Let u and v be two given functions of x, y, z, connected by an arbitrary function $\Phi(u, v) = 0$. By eliminating the arbitrary function Φ from $\Phi(u, v) = 0$, we obtain **Lagrange's Linear PDE**:

$$Pp + Qq = R$$

Example

Form the partial differential equation by eliminating the arbitrary function from

$$z = f(x^2 + y^2). \tag{1}$$

Solution:

Let

$$p = \frac{\partial z}{\partial x}, \qquad q = \frac{\partial z}{\partial y}.$$

Our objective is to eliminate the arbitrary function f and its derivative f'.

Step 1: Introduce a New Variable

From Equation (1), we have

$$z = f(x^2 + y^2).$$

Put

$$u = x^2 + y^2. \tag{2}$$

Therefore,

$$z = f(u). \tag{3}$$

Since u depends on both x and y, we apply the chain rule when taking partial derivatives.

Step 2: Differentiate with Respect to x

Taking the partial derivative of Equation (3) with respect to x, while keeping y constant, gives

$$\frac{\partial z}{\partial x} = f'(u)\frac{\partial u}{\partial x}.$$

From Equation (2),

$$u = x^2 + y^2.$$

Hence,

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x}(x^2 + y^2) = 2x + 0 = 2x.$$

Substituting, we obtain

$$\boxed{p = 2xf'(x^2 + y^2).} \tag{4}$$

Step 3: Differentiate with Respect to y

Similarly, taking the partial derivative of Equation (3) with respect to y, while keeping x constant, gives

$$\frac{\partial z}{\partial y} = f'(u)\frac{\partial u}{\partial y}.$$

Since

$$u = x^2 + y^2,$$

we obtain

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y}(x^2 + y^2) = 0 + 2y = 2y.$$

Therefore,

$$\boxed{q = 2yf'(x^2 + y^2).} \tag{5}$$

Step 4: Eliminate the Arbitrary Function

We now have two equations:

$$p = 2xf'(x^2 + y^2), \tag{4}$$
$$q = 2yf'(x^2 + y^2). \tag{5}$$

Multiply Equation (4) by y:

$$yp = 2xyf'(x^2 + y^2). \tag{6}$$

Multiply Equation (5) by x:

$$xq = 2xyf'(x^2 + y^2). \tag{7}$$

Observe that the right-hand sides of Equations (6) and (7) are identical.

Step 5: Obtain the Required PDE

Subtract Equation (7) from Equation (6):

$$yp - xq = 2xyf'(x^2 + y^2) - 2xyf'(x^2 + y^2).$$

The terms containing the arbitrary function cancel. Thus,

$$yp - xq = 0.$$

Recall that

$$p = \frac{\partial z}{\partial x}, \qquad q = \frac{\partial z}{\partial y}.$$

Therefore, the required PDE is

Final Answer

$$\boxed{y\frac{\partial z}{\partial x} - x\frac{\partial z}{\partial y} = 0.} \tag{8}$$

Solution of Lagrange's Linear PDE

Refer to Lagrange's auxiliary system and method of multipliers:

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

giving independent integrals $u(x, y, z) = c_1$ and $v(x, y, z) = c_2$, leading to the general solution $\Phi(u, v) = 0$.

Types of Solutions of PDEs

Solutions/Integrals of PDEs: A solution or integral of a PDE is a relation between the independent and dependent variables which satisfies the given PDE.
There are two distinct types of solutions of PDEs:

1. One type containing **arbitrary constants**.
2. Another type containing an **arbitrary function**.

Both types may serve as solutions to the same PDE.

Consider the equations:

$$z = ax + by \quad \text{— (1)} \qquad \text{and} \qquad z = xf\left(\frac{x}{y}\right) \quad \text{— (2)}$$

where a, b are arbitrary constants and f is an arbitrary function.

Classification of Solutions

Eliminating a, b from $z = ax + by$ and f from $z = xf(x/y)$ both yield the same PDE:

$$px + qy = z$$

Thus, $z = ax + by$ and $z = xf(x/y)$ are both solutions of $px + qy = z$.

Definitions:

- **Complete Integral:** A solution containing as many arbitrary constants as there are independent variables.
- **Particular Integral:** A solution obtained by assigning specific values to the arbitrary constants in a complete integral.
- **Singular Integral:** Derived via envelope conditions.

Method to Find the Singular Integral

Suppose

$$f(x, y, z, p, q) = 0 \quad \text{— (1)}$$

is a PDE whose complete integral is:

$$\phi(x, y, z, a, b) = 0 \quad \text{— (2)}$$

where a and b are arbitrary constants.
Differentiating $\phi(x, y, z, a, b) = 0$ partially with respect to a and b:

$$\frac{\partial \phi}{\partial a} = 0 \quad \text{— (3)}$$

$$\frac{\partial \phi}{\partial b} = 0 \quad \text{— (4)}$$

The eliminant of a and b from equations (2), (3), and (4), when it exists, is called the **Singular Integral** of (1).

General Integral & Direct Integration

To find the General Integral: In the complete integral $\phi(x, y, z, a, b) = 0$, assume one constant is a function of the other: $b = f(a)$.

$$\phi(x, y, z, a, f(a)) = 0 \quad \text{— (5)}$$

Differentiating partially with respect to a:

$$\frac{\partial \phi}{\partial a} + \frac{\partial \phi}{\partial b} f'(a) = 0 \quad \text{— (6)}$$

Eliminating a between (5) and (6) yields the **General Integral**.

Its total differential is

$$d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz + \frac{\partial \phi}{\partial a} da + \frac{\partial \phi}{\partial b} db.$$

Since we differentiate with respect to a, keeping x, y, z fixed,

$$dx = 0, \qquad dy = 0, \qquad dz = 0.$$

Therefore,

$$d\phi = \frac{\partial \phi}{\partial a} da + \frac{\partial \phi}{\partial b} db.$$

Recall that

$$b = f(a).$$

Differentiating with respect to a,

$$\frac{db}{da} = f'(a).$$

Hence,

$$db = f'(a)\, da.$$

Substitute this expression for db into $d\phi$:

$$d\phi = \frac{\partial \phi}{\partial a} da + \frac{\partial \phi}{\partial b} f'(a)\, da.$$

Factor out da:

$$d\phi = \left[\frac{\partial \phi}{\partial a} + \frac{\partial \phi}{\partial b} f'(a) \right] da.$$

Dividing by da, we obtain

$$\frac{d\phi}{da} = \frac{\partial \phi}{\partial a} + \frac{\partial \phi}{\partial b} f'(a).$$

Solution of PDEs by Direct / Successive Integration:

A PDE can be solved by successive integration in cases where derivatives appear directly.

Example 1: Solve $\dfrac{\partial^2 z}{\partial x \partial y} = \sin x$

Integrating with respect to x:

$$\frac{\partial z}{\partial y} = -\cos x + f(y) \quad \text{where } f(y) \text{ is arbitrary}$$

Integrating with respect to y:

$$z = -y\cos x + F(y) + G(x) \quad \text{where } F(y) = \int f(y)dy$$

Example 2: Solve $\dfrac{\partial^2 z}{\partial x^2} = xy$

Integrating with respect to x:

$$\frac{\partial z}{\partial x} = \frac{x^2}{2} y + f(y)$$

Integrating with respect to x again:

$$z = \frac{x^3}{6} y + x f(y) + G(y)$$

Methods to solve the first order partial differential equations

The partial differential equation of the first order can be written as $F(x, y, z, p, q) = 0$ where $p = \frac{\partial z}{\partial x}$ and $q = \frac{\partial z}{\partial y}$. We shall see some standard forms of such equations and solve them by special methods.

Type I: $F(p, q) = 0$

Here the PDE contains p and q only.
Assume a trial complete integral:

$$z = ax + by + c$$

Then:

$$p = \frac{\partial z}{\partial x} = a, \quad q = \frac{\partial z}{\partial y} = b$$

Substituting $p = a$ and $q = b$ into $F(p, q) = 0$ yields the algebraic relation:

$$F(a, b) = 0$$

Hence the solution is $z = ax + by + c$, where $F(a, b) = 0$.

Complete Integral

Solving for b from $F(a, b) = 0$, we get $b = \phi(a)$. Then

$$\boxed{z = ax + \phi(a)y + c.} \tag{1}$$

This is the **complete integral** of the given equation since it contains two arbitrary constants, a and c.

Singular Integral: Envelope Conditions

The singular integral would be obtained by eliminating a and c from the complete integral and its envelope conditions:

$$z = ax + \phi(a)y + c,$$
$$0 = \frac{\partial}{\partial a}\left(ax + \phi(a)y + c\right) = x + \phi'(a)y,$$
$$0 = \frac{\partial}{\partial c}\left(ax + \phi(a)y + c\right) = 1.$$

The last equation, $0 = 1$, is impossible. Hence there is **no singular integral obtained as the envelope of this complete integral**.

General Integral: Introduce an Arbitrary Function

To find the general integral, put $c = f(a)$, f being arbitrary. Then Equation (1) becomes

$$z = ax + y\phi(a) + f(a). \tag{2}$$

Differentiating partially with respect to the parameter a, while holding x, y, z fixed for the envelope condition, we get

$$\begin{aligned} 0 &= \frac{\partial}{\partial a}\left(ax + y\phi(a) + f(a)\right) \\ &= x + y\phi'(a) + f'(a). \end{aligned}$$

Thus

$$\boxed{0 = x + y\phi'(a) + f'(a).} \tag{3}$$

Eliminate the Parameter a

Eliminating a between Equations (2) and (3), we get the **general integral**:

$$\boxed{\begin{cases} z = ax + y\phi(a) + f(a), \\ 0 = x + y\phi'(a) + f'(a). \end{cases}}$$

Here f is an arbitrary differentiable function. When the second equation can be solved for $a = a(x, y)$, substitute it into the first to obtain $z = z(x, y)$.

Example on Type I: $\sqrt{p} + \sqrt{q} = 1$

Solve: $\sqrt{p} + \sqrt{q} = 1$
This is of the form $F(p, q) = 0$.
Set complete integral: $z = ax + by + c$ with $\sqrt{a} + \sqrt{b} = 1$.
Solving for b:

$$\sqrt{b} = 1 - \sqrt{a} \implies b = (1 - \sqrt{a})^2$$

Thus, the Complete Solution is:

$$z = ax + (1 - \sqrt{a})^2 y + c \quad \text{— (*)}$$

Singular Solution check: Differentiating (*) with respect to c:

$$\frac{\partial z}{\partial c} = 0 \implies 1 = 0 \quad \text{(Impossible)}$$

Hence, there is no singular solution.

Type I Example: Finding the General Solution

For finding the general solution, set $c = f(a)$, where f is an arbitrary function:

$$z = ax + (1 - \sqrt{a})^2 y + f(a) \quad \text{— (**)}$$

Differentiating (**) partially with respect to a:

$$x + 2(1 - \sqrt{a}) \left(-\frac{1}{2\sqrt{a}} \right) y + f'(a) = 0$$

$$x - \frac{1 - \sqrt{a}}{\sqrt{a}} y + f'(a) = 0 \quad \text{— (***)}$$

Rearranging (***):

$$x + f'(a) = \frac{1 - \sqrt{a}}{\sqrt{a}} y = \left(\frac{1}{\sqrt{a}} - 1 \right) y$$

$$x + y + f'(a) = \frac{y}{\sqrt{a}}$$

Type I Example: Parametric Form of General Solution

Solving for $\sqrt{a}$ from $x + y + f'(a) = \dfrac{y}{\sqrt{a}}$:

$$\sqrt{a} = \frac{y}{x + y + f'(a)}$$

Squaring both sides gives a:

$$a = \frac{y^2}{(x + y + f'(a))^2} \quad \text{— (****)}$$

The general integral is given by eliminating a between (**) and (****).

Special Choice for $f(a)$: Let $f(a) = k$ (a constant), then $f'(a) = 0$.

Type I Example: Constant $f(a) = k$

When $f(a) = k$ (so $f'(a) = 0$):

$$\sqrt{a} = \frac{y}{x+y} \implies a = \frac{y^2}{(x+y)^2}$$

Then:

$$1 - \sqrt{a} = 1 - \frac{y}{x+y} = \frac{x}{x+y}$$

$$(1 - \sqrt{a})^2 = \frac{x^2}{(x+y)^2}$$

Substitute a and $(1 - \sqrt{a})^2$ into $z = ax + (1 - \sqrt{a})^2 y + k$:

$$z = \left(\frac{y^2}{(x+y)^2}\right) x + \left(\frac{x^2}{(x+y)^2}\right) y + k = \frac{xy^2 + x^2 y}{(x+y)^2} + k$$

Type I Example: Verification

Simplifying z:

$$z = \frac{xy(x+y)}{(x+y)^2} + k = \frac{xy}{x+y} + k$$

Verification of Partial Derivatives:

$$p = z_x = \frac{y(x+y) - 1(xy)}{(x+y)^2} = \frac{y^2}{(x+y)^2} \implies \sqrt{p} = \frac{y}{x+y}$$

$$q = z_y = \frac{x(x+y) - 1(xy)}{(x+y)^2} = \frac{x^2}{(x+y)^2} \implies \sqrt{q} = \frac{x}{x+y}$$

$$\sqrt{p} + \sqrt{q} = \frac{y}{x+y} + \frac{x}{x+y} = \frac{x+y}{x+y} = 1 \quad \checkmark$$

Type I Example: Envelope Analysis

General Form parameterized by arbitrary $f(a)$:

$$z = \frac{xy^2 + y(x + f'(a))^2}{(x + y + f'(a))^2} + f(a), \quad \text{where } \sqrt{a} = \frac{y}{x + y + f'(a)}$$

Investigating Singular Solution via Envelope of $z = ax + (1 - \sqrt{a})^2 y + c$:

Envelope conditions require:

$$\frac{\partial z}{\partial a} = 0 \implies x - \frac{1 - \sqrt{a}}{\sqrt{a}} y = 0 \implies \sqrt{a} = \frac{y}{x + y}$$

$$\frac{\partial z}{\partial c} = 0 \implies 1 = 0 \quad \text{(Impossible)}$$

Hence, no singular solution exists in the usual envelope sense.

Particular Integrals

Complete Integral: $z = ax + (1 - \sqrt{a})^2 y + c$

Particular Integral 1: Choose $a = \frac{1}{4}$ and $c = 2$:

$$\sqrt{a} = \frac{1}{2} \implies (1 - \sqrt{a})^2 = \frac{1}{4}$$

$$z = \frac{1}{4}x + \frac{1}{4}y + 2$$

Particular Integral 2: Choosing a non-zero function $f(a) = \sqrt{a}$:

$$z = ax + (1 - \sqrt{a})^2 y + \sqrt{a}$$

Choosing $f(a) = \sqrt{a}$ yields $f'(a) = \frac{1}{2\sqrt{a}}$, leading to:

$$z = \frac{xy + y - 1/4}{2(x + y)}$$

This provides another integrated solution.

Type II: Clairaut's Form: $z = px + qy + f(p, q)$

Suppose that the given equation is of the form

$$z = px + qy + f(p, q) \quad \text{— (1)}$$

The **Complete Solution** is:

$$z = ax + by + f(a, b) \quad \text{— (2)}$$

where a and b are arbitrary constants.

Clairaut's Form: Singular and General Integrals

Differentiating (2) partially with respect to a and b:

$$x + \frac{\partial f}{\partial a} = 0 \quad \text{— (3)}$$

$$y + \frac{\partial f}{\partial b} = 0 \quad \text{— (4)}$$

Eliminating a and b from (2), (3), and (4) gives the **Singular Integral**.

General Integral: Set $b = \phi(a)$ in (2):

$$z = ax + \phi(a)y + f(a, \phi(a)) \quad \text{— (5)}$$

Differentiating with respect to a:

$$x + y\phi'(a) + f'(a) = 0 \quad \text{— (6)}$$

Eliminating a between (5) and (6) gives the **General Integral**.

Example on Clairaut's Form: $z = px + qy + p^2q^2$

Ex: Solve $z = px + qy + p^2q^2$

Complete Solution:

$$z = ax + by + a^2b^2$$

Differentiate with respect to a and b to find the envelope conditions:

$$x + 2ab^2 = 0 \implies x = -2ab^2$$

$$y + 2a^2b = 0 \implies y = -2a^2b$$

Divide x and y by b and a respectively:

$$\frac{x}{b} = -2ab, \quad \frac{y}{a} = -2ab \implies \frac{x}{b} = \frac{y}{a} = -2ab$$

Clairaut's Form Example: Eliminating Constants

Multiplying x and y:

$$x \cdot y = (-2ab^2)(-2a^2b) = 4a^3b^3 = 4(ab)^3$$

$$(ab)^3 = \frac{xy}{4} \implies ab = \left(\frac{xy}{4}\right)^{1/3}$$

Now express ax and by:

$$ax = -2a^2b^2, \quad by = -2a^2b^2$$

Substitute into complete solution $z = ax + by + a^2b^2$:

$$z = -2a^2b^2 - 2a^2b^2 + a^2b^2 = -3a^2b^2 = -3(ab)^2$$

Clairaut's Form Example: Singular Solution

Since $ab = \left(\frac{xy}{4}\right)^{1/3}$, we have:

$$(ab)^2 = \left(\frac{xy}{4}\right)^{2/3}$$

Substitute $(ab)^2$ into $z = -3(ab)^2$:

$$z = -3\left(\frac{xy}{4}\right)^{2/3} = -3\sqrt[3]{\frac{x^2y^2}{16}}$$

Implicit Form: Cubing $z = -3(ab)^2$:

$$z^3 = -27a^6b^6$$

Since $xy = 4a^3b^3 \implies x^2y^2 = 16a^6b^6 \implies a^6b^6 = \frac{x^2y^2}{16}$.

Substituting $a^6 b^6$ into z^3:

$$z^3 = -\frac{27}{16} x^2 y^2 \implies 16z^3 + 27x^2 y^2 = 0$$

Type III:a) $F(z, p, q) = 0$

Assume trial solution $z = f(u)$, where $u = x + ay$ (a constant).

$$p = \frac{\partial z}{\partial x} = \frac{dz}{du}\frac{\partial u}{\partial x} = \frac{dz}{du} \cdot 1 = \frac{dz}{du}$$

$$q = \frac{\partial z}{\partial y} = \frac{dz}{du}\frac{\partial u}{\partial y} = \frac{dz}{du} \cdot a = a\frac{dz}{du}$$

Substitute $p = \frac{dz}{du}$ and $q = a\frac{dz}{du}$ into $F(z, p, q) = 0$:

$$F\left(z, \frac{dz}{du}, a\frac{dz}{du}\right) = 0$$

, which is an ordinary differential equation of the first order.

Type III Method Procedure

Solving

$$F\left(z, \frac{dz}{du}, a\frac{dz}{du}\right) = 0$$

for $\frac{dz}{du}$, we obtain

$$\frac{dz}{du} = \phi(z, a) \implies \frac{dz}{\phi(z, a)} = du$$

Integrating both sides:

$$\int \frac{dz}{\phi(z, a)} = \int du \implies f(z, a) = u + c$$

Substitute $u = x + ay$:

$$f(z, a) = x + ay + c$$

This is the complete integral.

Example on Type III: $z = p^2 + q^2$

Solve: $z = p^2 + q^2$

Let $u = x + ay$, so $p = \dfrac{dz}{du}$ and $q = a\dfrac{dz}{du}$.

Substitute p and q into $z = p^2 + q^2$:

$$z = \left(\frac{dz}{du}\right)^2 + \left(a\frac{dz}{du}\right)^2 = (1 + a^2)\left(\frac{dz}{du}\right)^2$$

Solving for $\dfrac{dz}{du}$:

$$\left(\frac{dz}{du}\right)^2 = \frac{z}{1 + a^2} \implies \frac{dz}{du} = \frac{\sqrt{z}}{\sqrt{1 + a^2}}$$

Separating variables:

$$\frac{dz}{\sqrt{z}} = \frac{du}{\sqrt{1 + a^2}}$$

Type III Example Completion & Type IV Intro

Integrating both sides:

$$\int \frac{dz}{\sqrt{z}} = \int \frac{du}{\sqrt{1+a^2}} \implies 2\sqrt{z} = \frac{u}{\sqrt{1+a^2}} + c$$

Substituting $u = x + ay$:

$$2\sqrt{z} = \frac{x+ay}{\sqrt{1+a^2}} + c$$

This is the complete solution.

Type III b) $F(x, p, q) = 0$
Since $dz = p\,dx + q\,dy$, assume $q = a$. Solve $F(x, p, a) = 0$ for p: $p = \phi(x, a)$.

$$dz = \phi(x,a)dx + ady \implies z = \int \phi(x,a)dx + ay + c$$

Example on $F(x, p, q) = 0$

Ex 1: Solve $p + q - x = 0$
Assume $q = a$, then $p + a - x = 0 \implies p = x - a$.

$$dz = p\,dx + q\,dy = (x - a)dx + a\,dy$$

Integrating:

$$z = \int (x - a)dx + \int a\,dy \implies z = \frac{x^2}{2} - ax + ay + c$$

Type III c) $F(y, p, q) = 0$**:** Assume $p = a$, then solve $F(y, a, q) = 0$ for $q = \phi(y, a)$.

$$dz = a\,dx + \phi(y, a)dy \implies z = ax + \int \phi(y, a)dy + c$$

Example on $F(y, p, q) = 0$

Ex 2: Solve $p + q - y = 0$
Assume $p = a$, then:

$$a + q - y = 0 \implies q = y - a$$

Thus $f(y, a) = y - a$.
Substitute $p = a$ and $q = y - a$ into $dz = p\,dx + q\,dy$:

$$dz = a\,dx + (y - a)dy$$

Integrating both sides:

$$\int dz = \int a\,dx + \int (y - a)dy$$

$$z = ax + \frac{y^2}{2} - ay + c$$

This is the complete integral, where $f(y, a) = \frac{y^2}{2} - ay$.